

## ***Multivitamin-multimineral supplementation and mortality: a meta-analysis of randomized controlled trials.***

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BACKGROUND: Multivitamins are the most commonly used supplement in the developed world. Recent epidemiologic findings suggest that multivitamin use increases the risk of mortality. OBJECTIVE: We aimed to determine whether multivitamin-multimineral treatment, used for primary or secondary prevention, increases the risk of mortality in independently living adults. DESIGN: We performed a meta-analysis of randomized controlled trials. Multiple electronic databases were systematically searched from March to October 2012. Randomized controlled primary or secondary prevention trials were considered for inclusion. Eligible trials investigated daily multivitamin-multimineral supplementation for  $\geq 1$  y. Cohorts described as institutionalized or as having terminal illness (tertiary prevention) were excluded. The number of deaths and the sample size of each study arm were extracted independently by 2 researchers. Twenty-one articles were included in the analysis, which generated a total pooled sample of 91,074 people and 8794 deaths. These trials were pooled in a meta-analysis, and the outcomes were expressed as RRs and 95% CIs. RESULTS: The average age of the pooled sample was 62 y, and the average duration of supplementation was 43 mo. Across all studies, no effect of multivitamin-multimineral treatment on all-cause mortality (RR: 0.98; 95% CI: 0.94, 1.02) was observed. There was a trend for a reduced risk of all-cause mortality across primary prevention trials (RR: 0.94; 95% CI: 0.89, 1.00). Multivitamin-multimineral treatment had no effect on mortality due to vascular causes (RR: 1.01; 95% CI: 0.93, 1.09) or cancer (RR: 0.96; 95% CI: 0.88, 1.04). No statistical evidence of heterogeneity or publication bias was observed. CONCLUSION: Multivitamin-multimineral treatment has no effect on mortality risk.